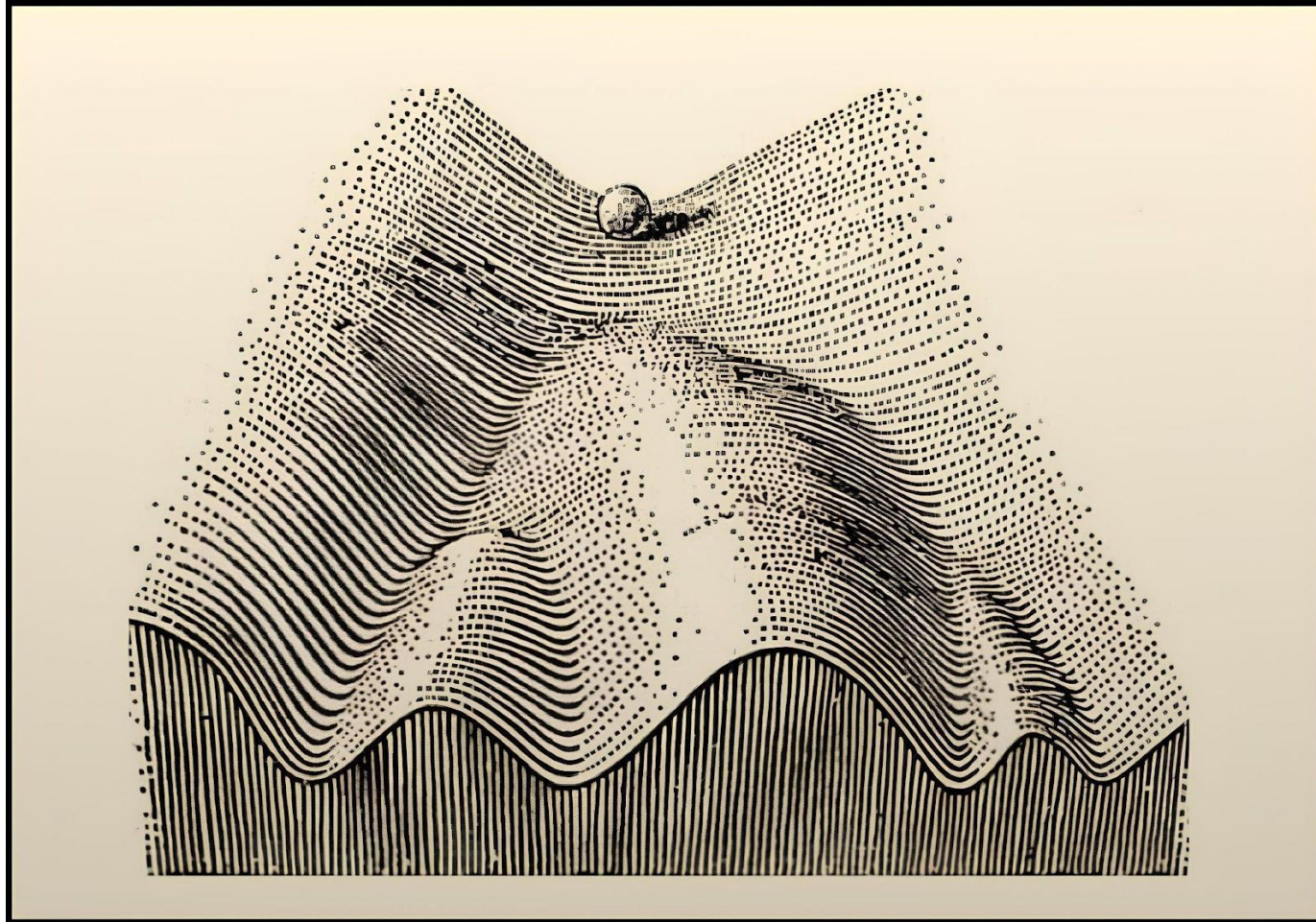
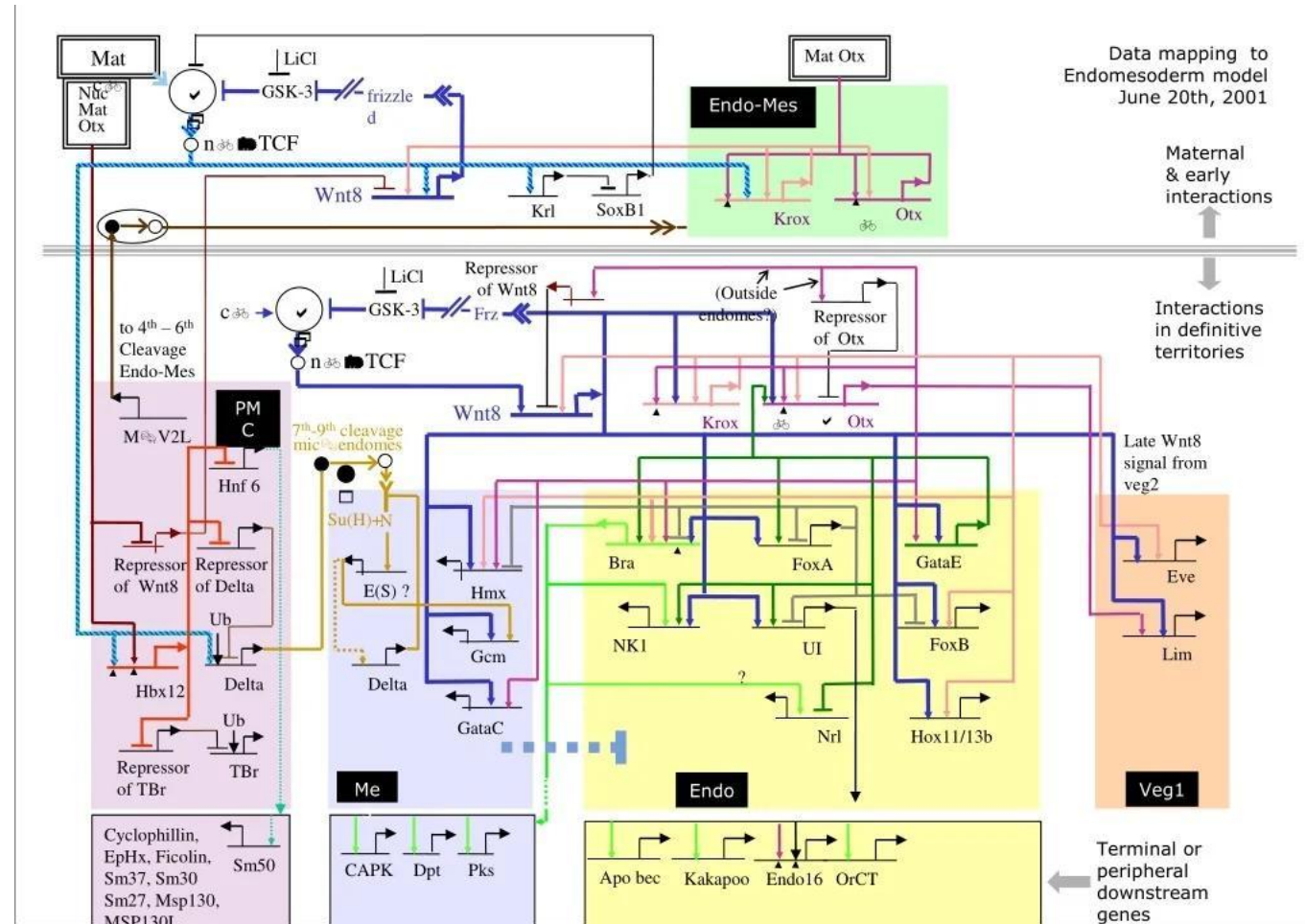


How can the same genome generate many stable cell identities?

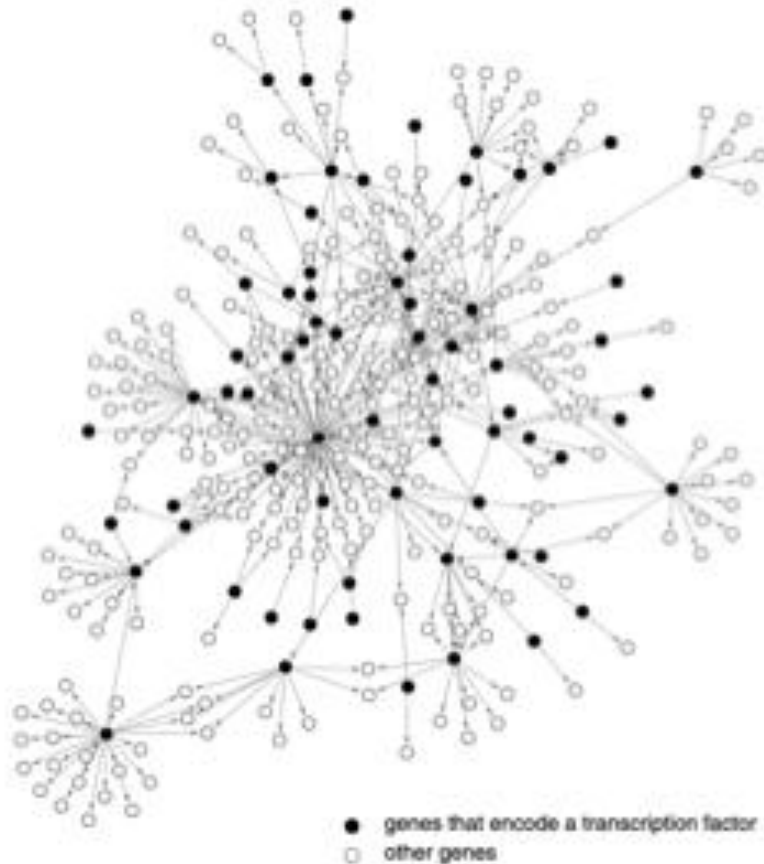


Adapted from Waddington, C.H. (1957), *The Strategy of the Genes*, Fig. 4.

Gene regulatory networks provide a mechanistic language for cell-fate decisions

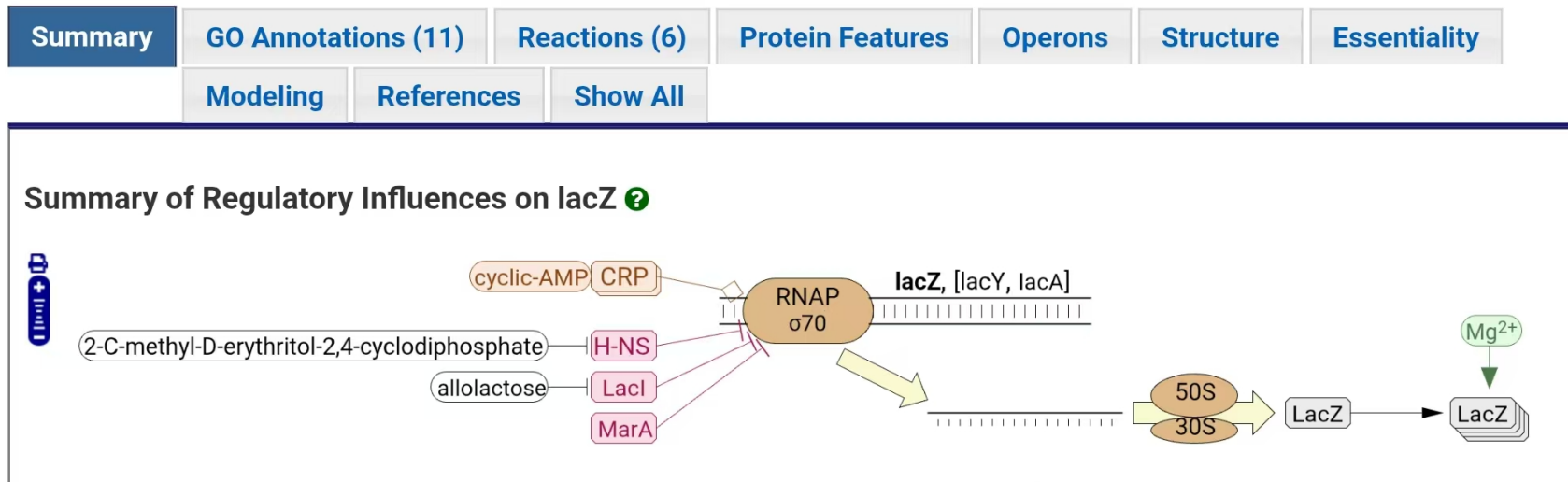


Do cells with the same genotype and environment have the same phenotypes?

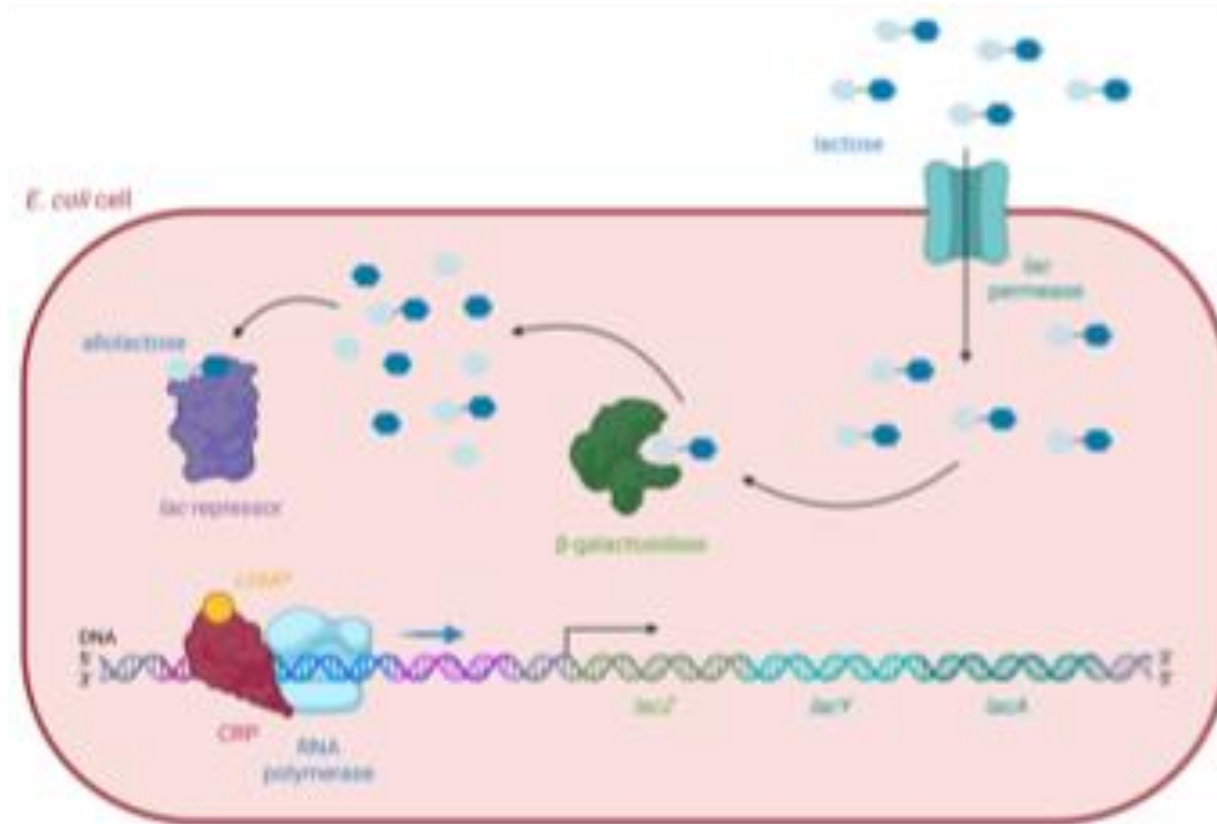


$$\frac{dx_i}{dt} = \beta_i(\mathbf{x}; \mathbf{s}) - [\mu(\mathbf{x}; \mathbf{s}) + \gamma_i] x_i$$

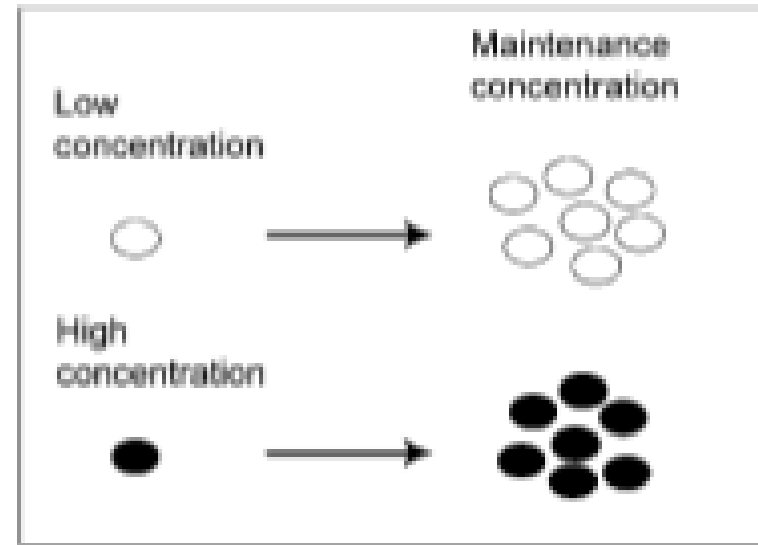
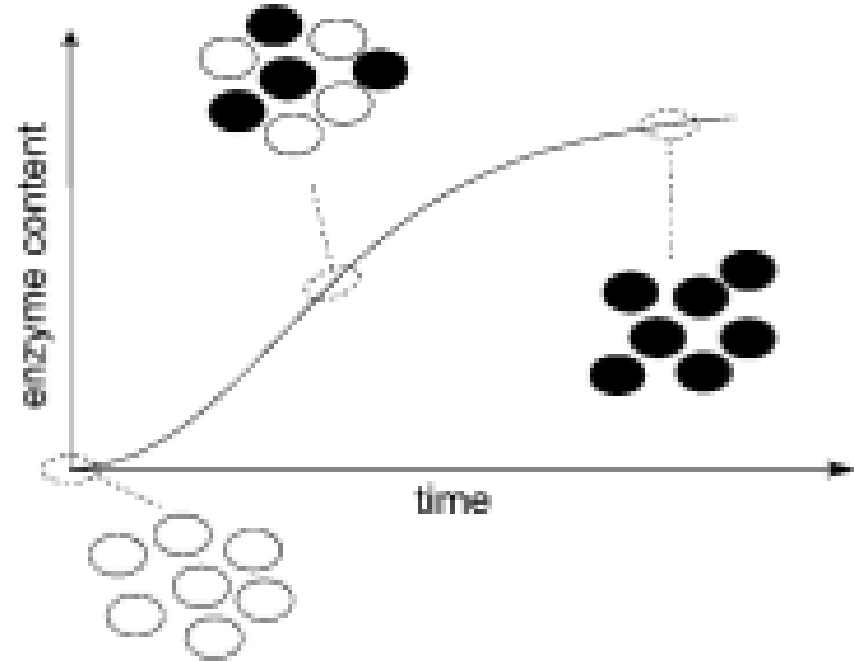
The lac operon is one of the best-studied gene regulatory systems



In *E. coli*, lactose induces the lac operon through allolactose

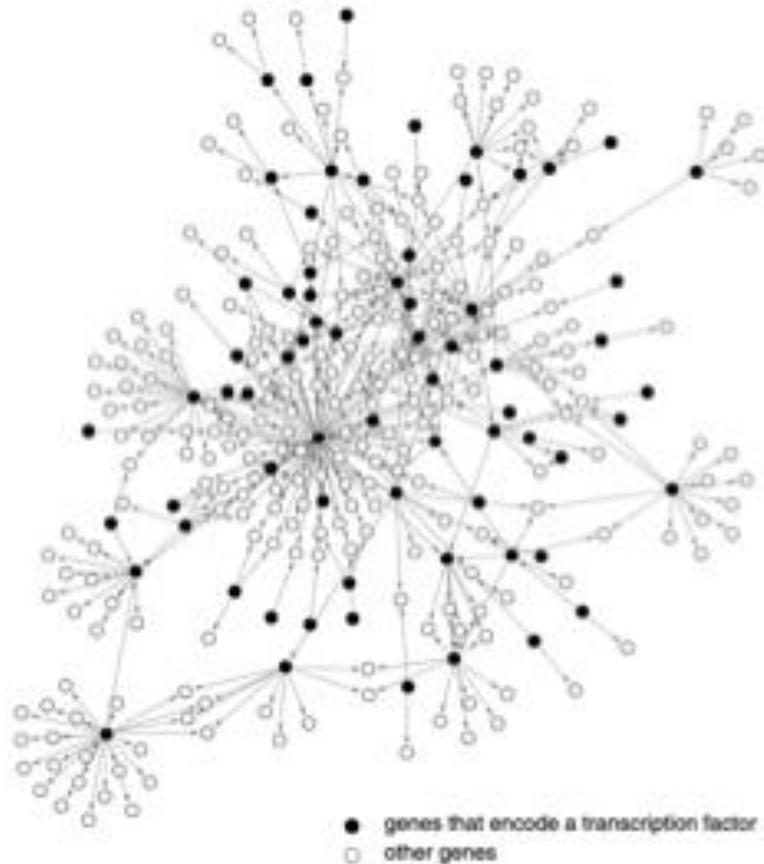


In 1957, Novick and Weiner found something remarkable in the lac operon



Novick & Weiner, PNAS (1957)
Vilar, Guet & Leibler, JCB (2003)

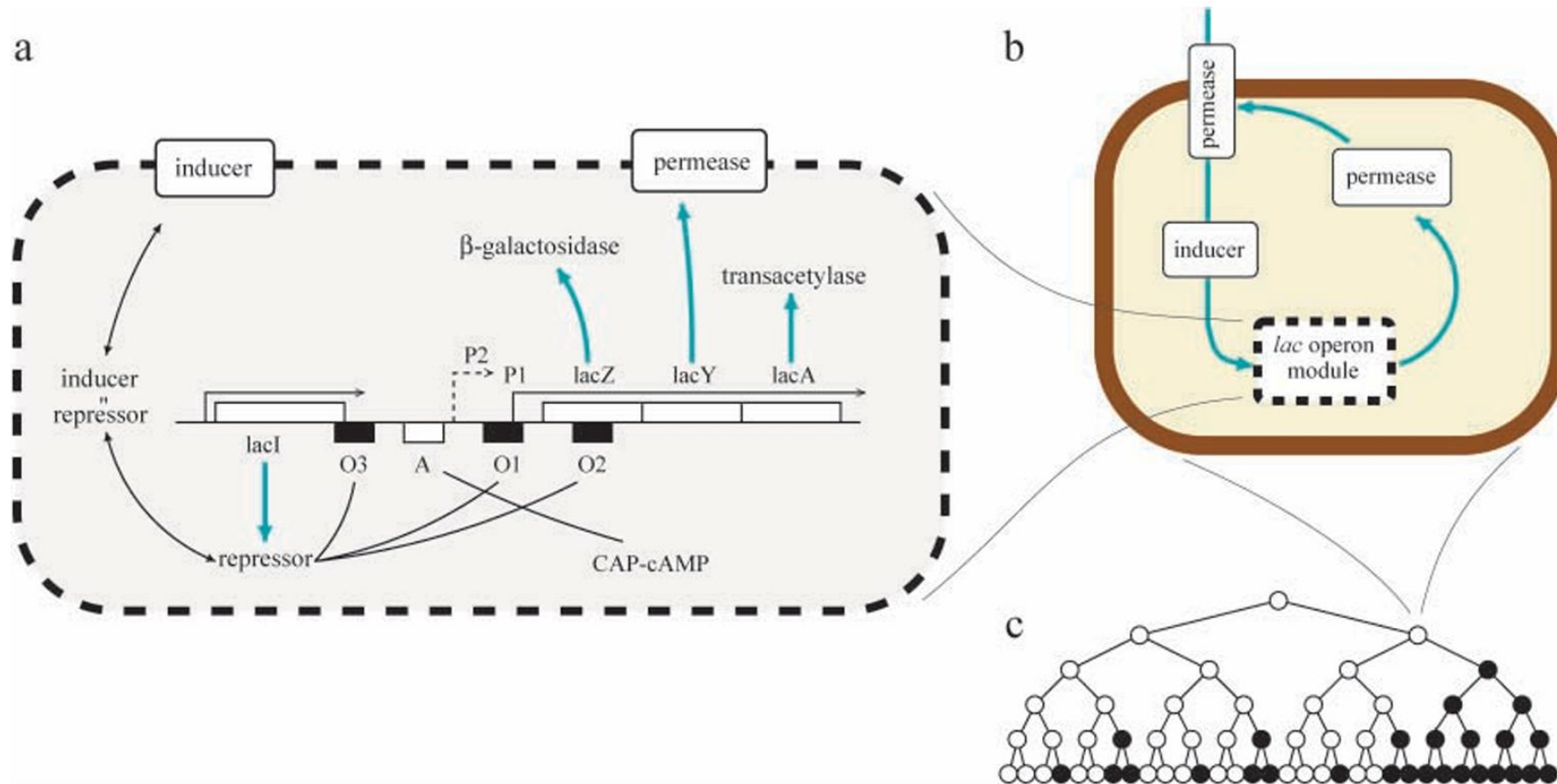
How can a cell remember longer than the molecules that make up its regulatory network?



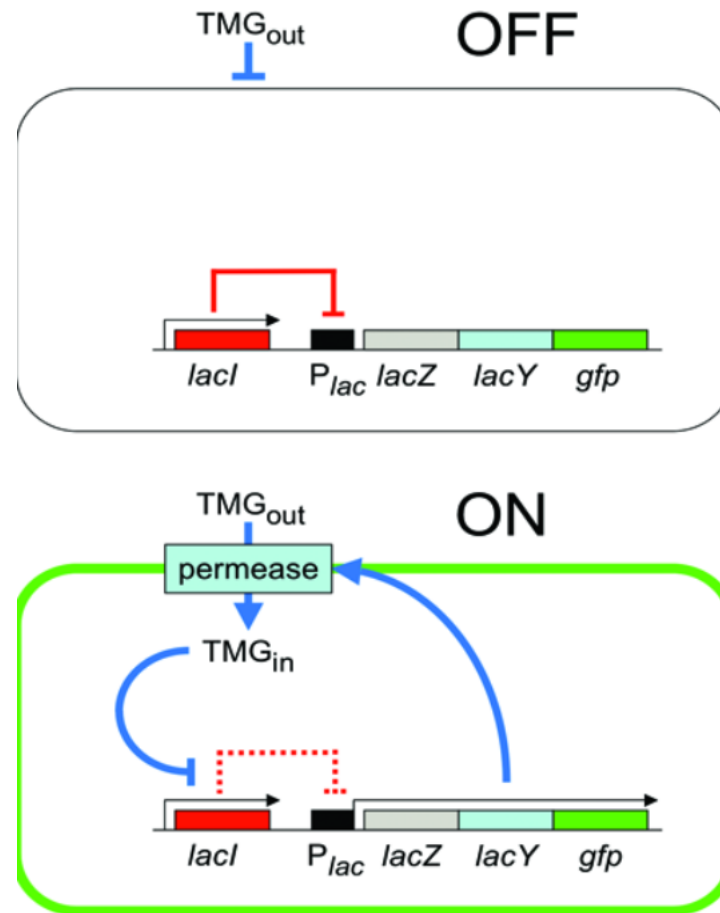
$$\frac{dx_i}{dt} = \beta_i(\mathbf{x}; \mathbf{s}) - [\mu(\mathbf{x}; \mathbf{s}) + \gamma_i] x_i$$

Modern single-cell imaging lets us watch epigenetic memory and all-or-none switching directly

Positive feedback provides a possible mechanism for lac-operon memory

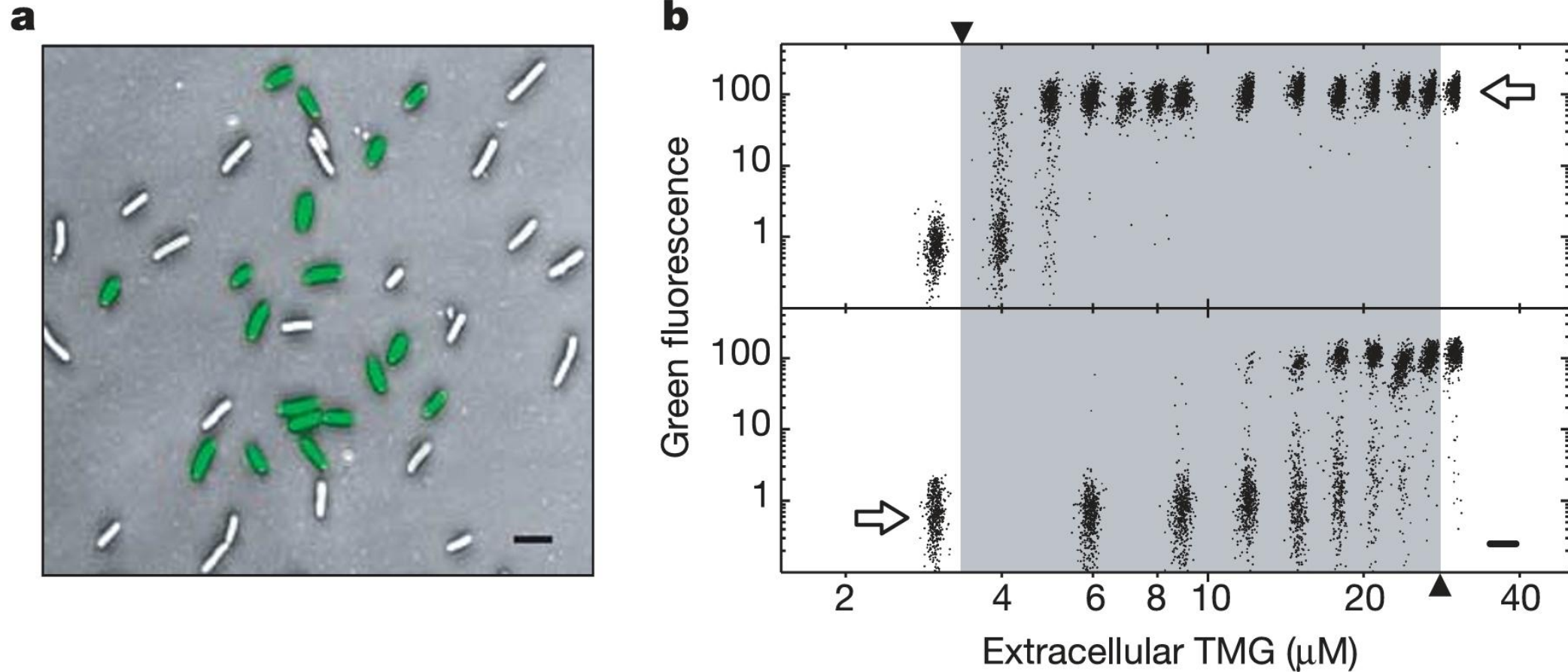


TMG isolates the regulatory feedback from lactose metabolism

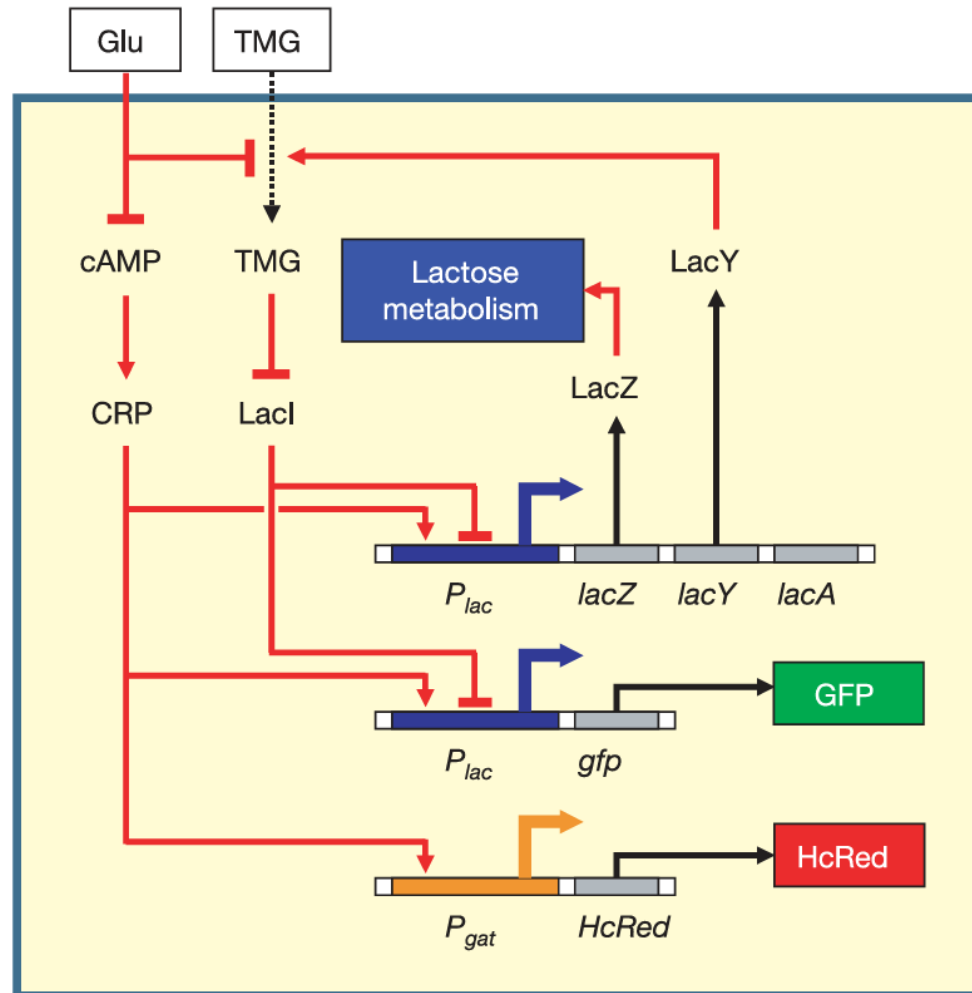


Gordon et al., PLoS Biol. (2009)

Ozbudak et al. revisited and quantified the Novick–Weiner phenomenon at the single-cell level

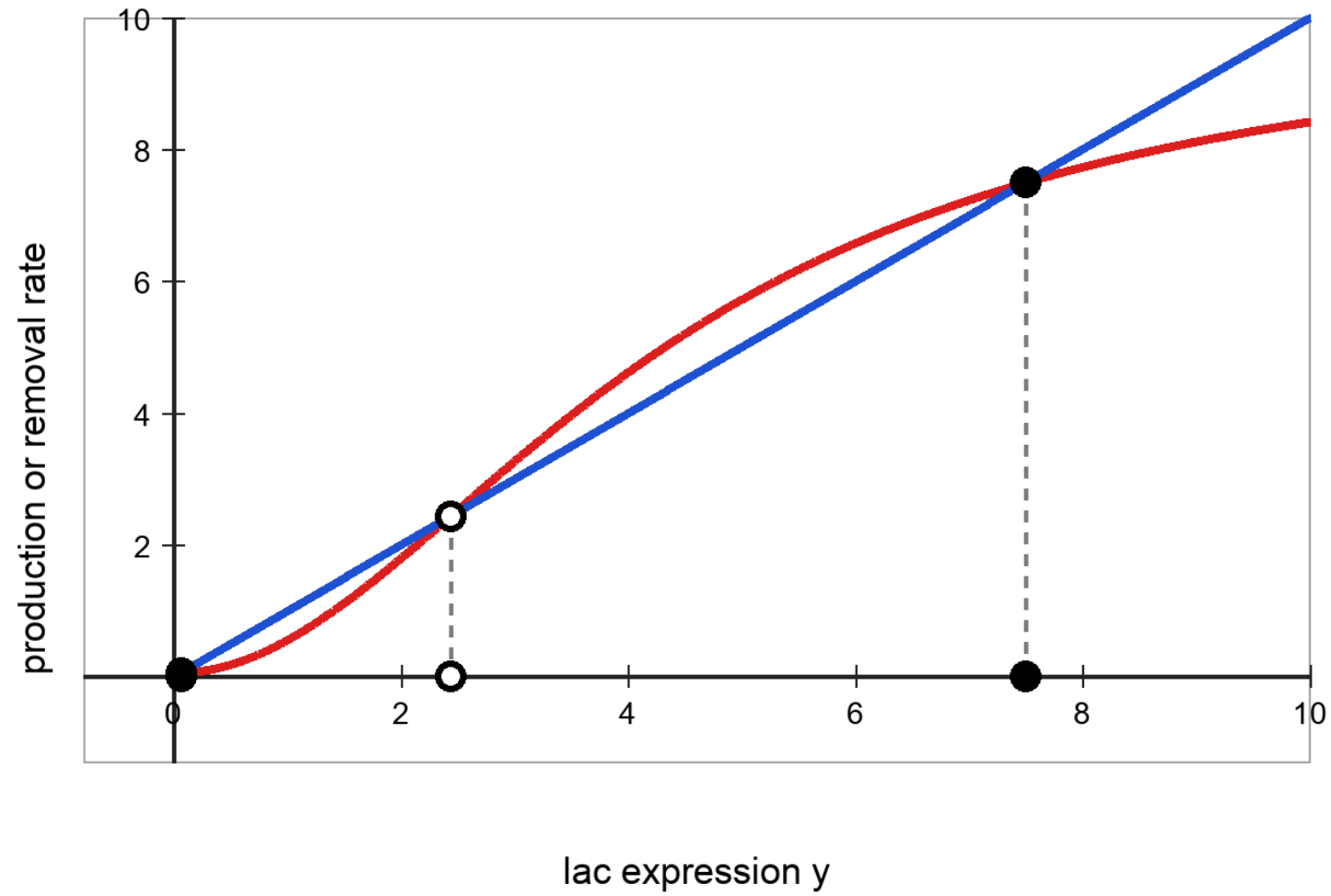


What is the minimal circuit needed to explain this behavior?



ON threshold $\beta = 3.000$

— production rate — removal rate

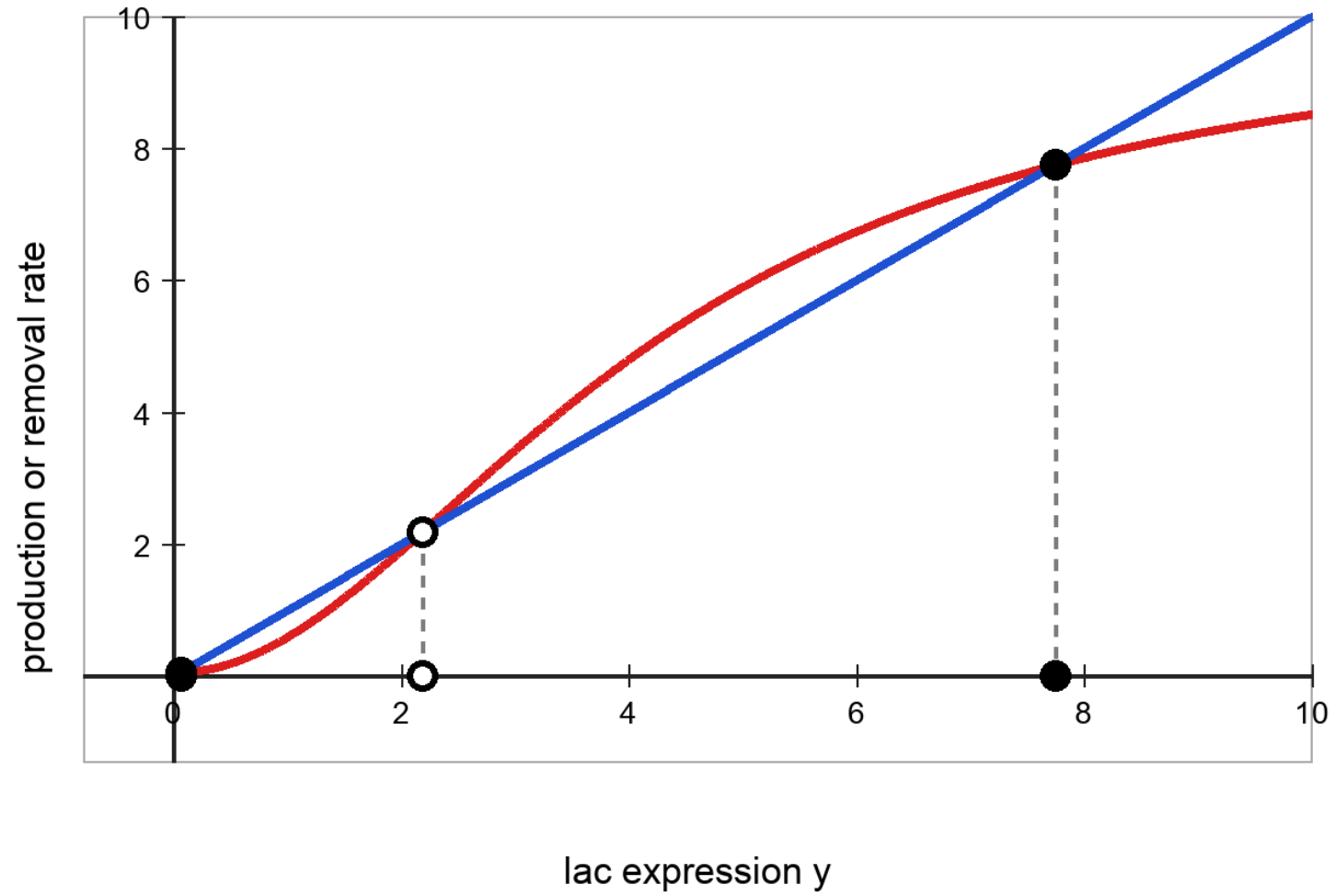


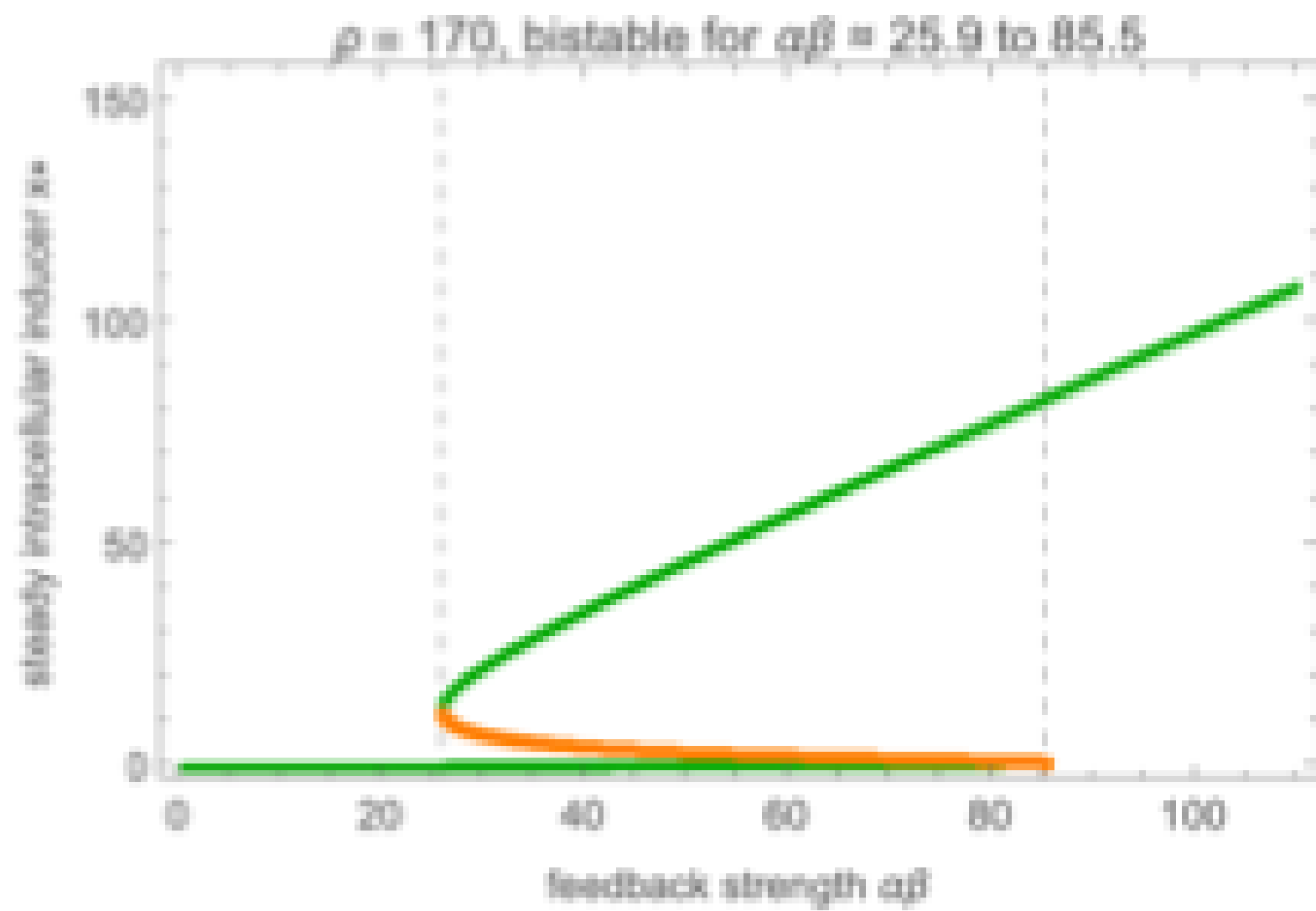
OFF threshold

beta = 3.111

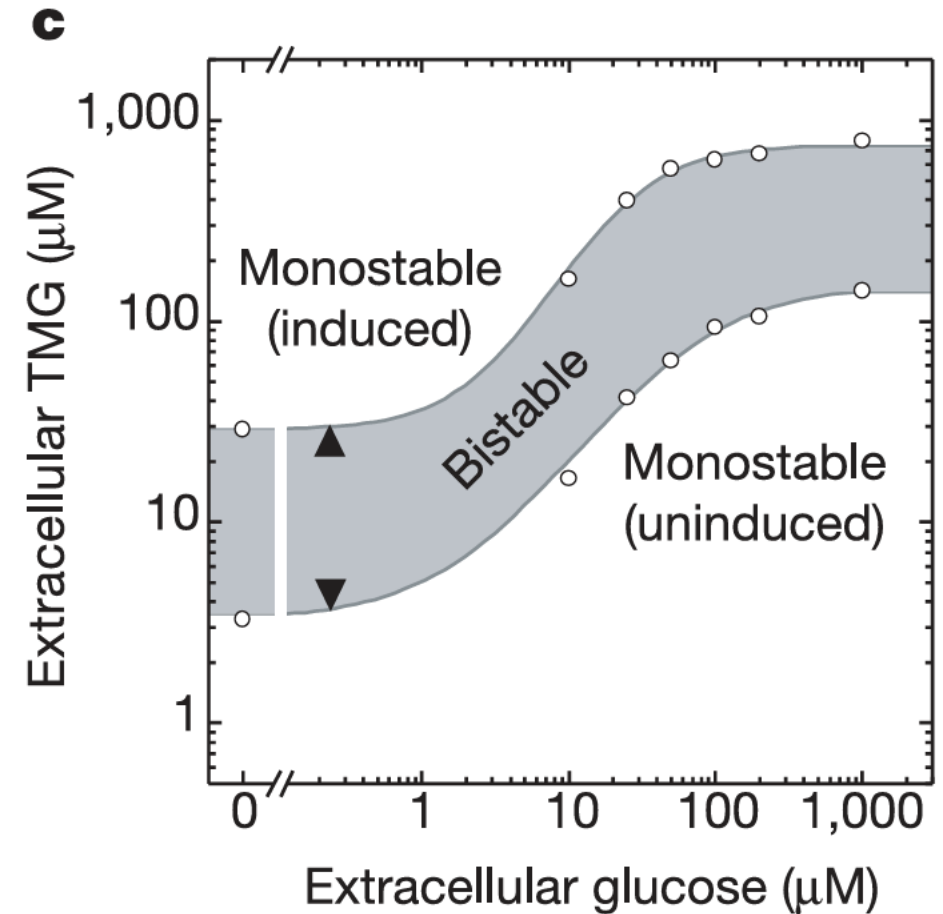
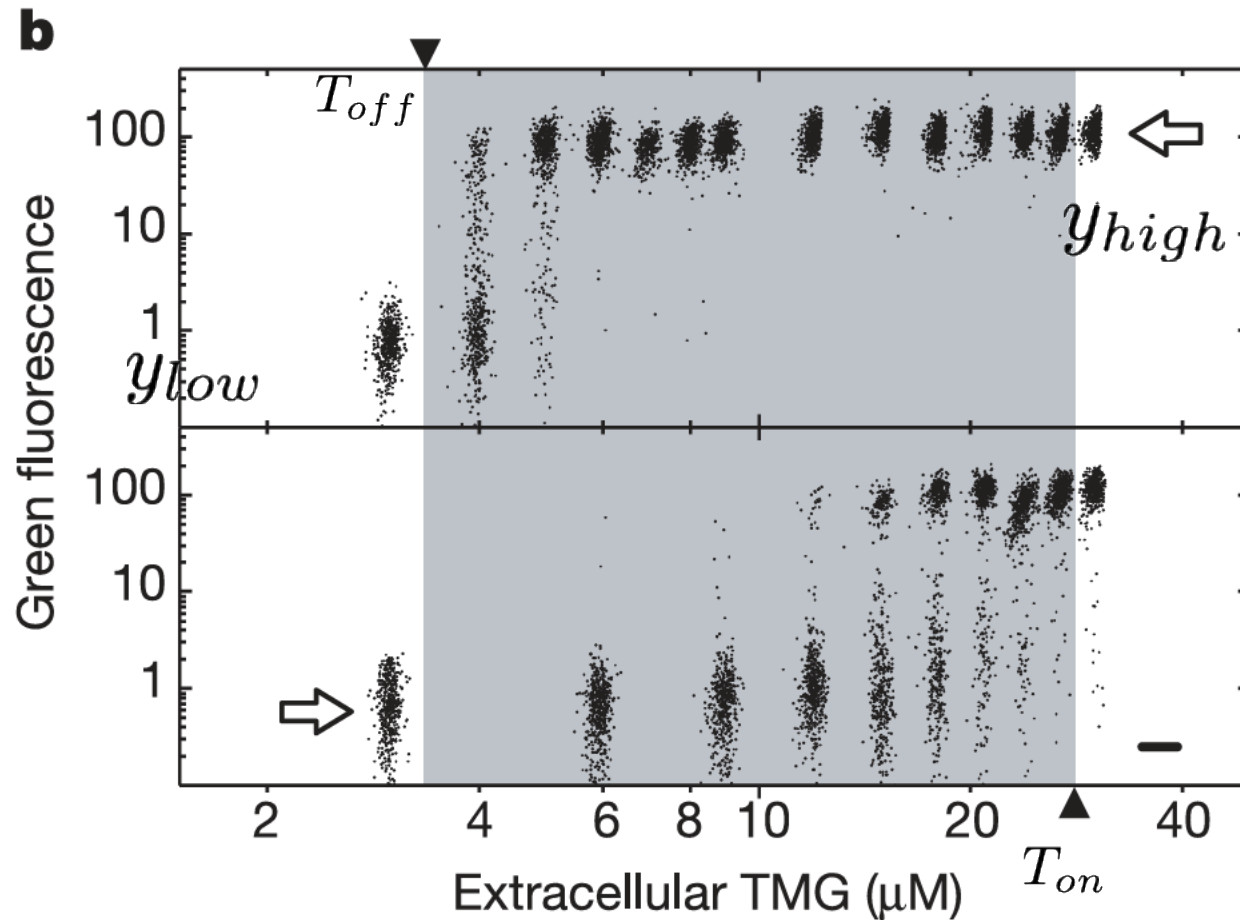
— production rate

— removal rate

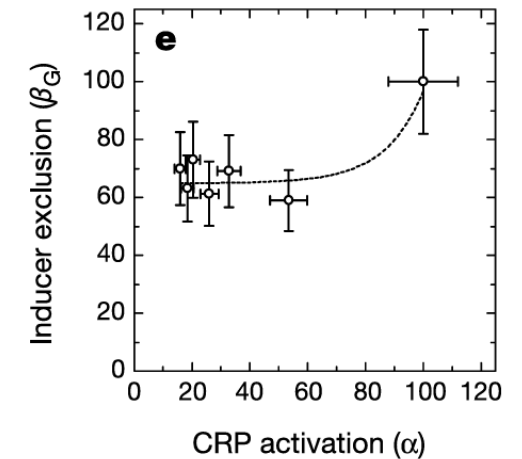
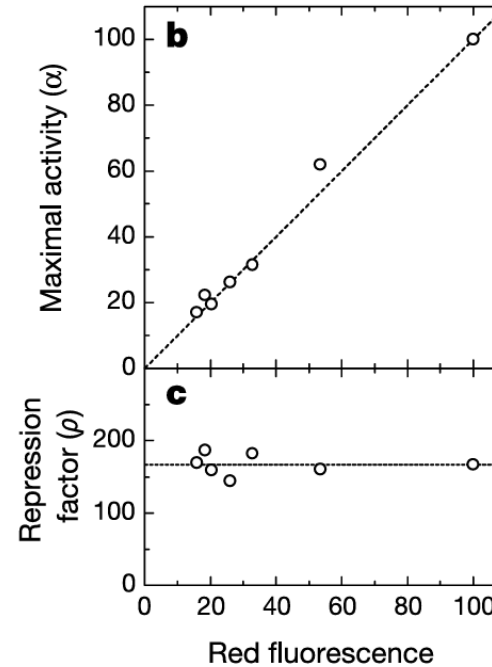
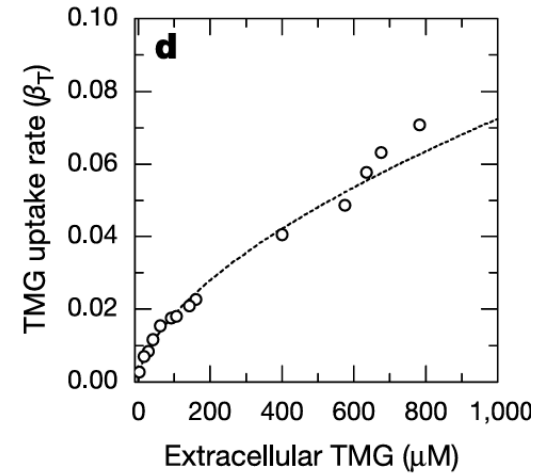
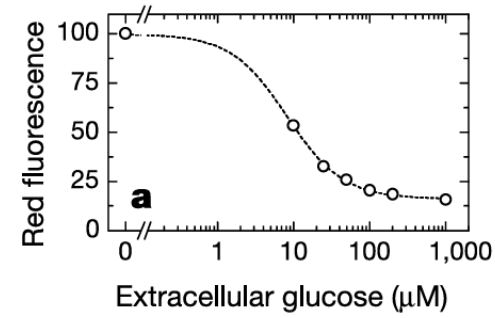
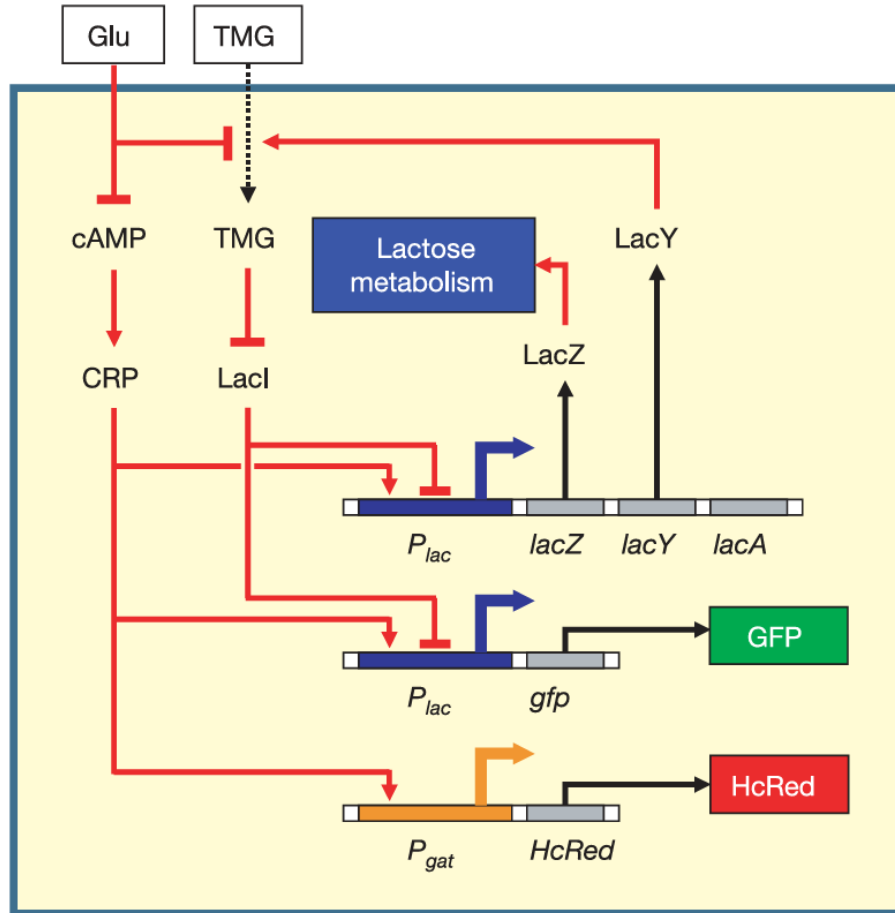




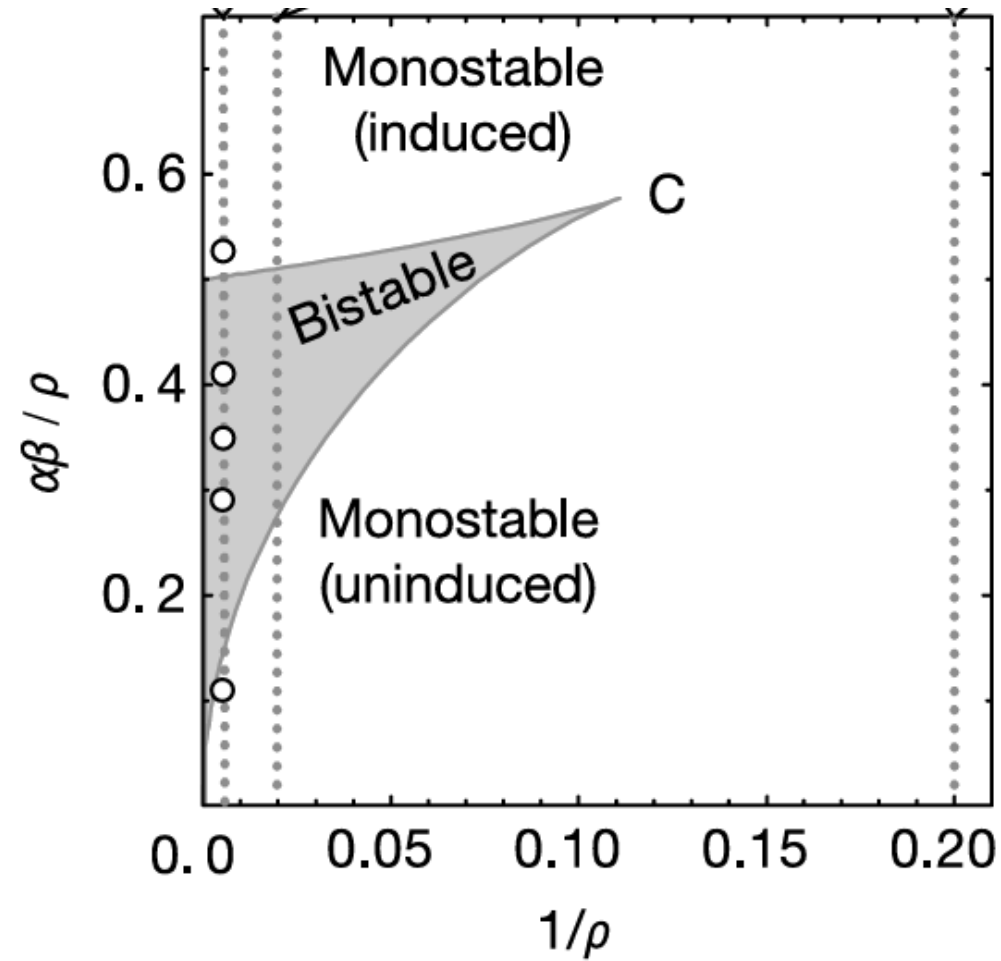
The two switching thresholds define a region of bistability in environmental space



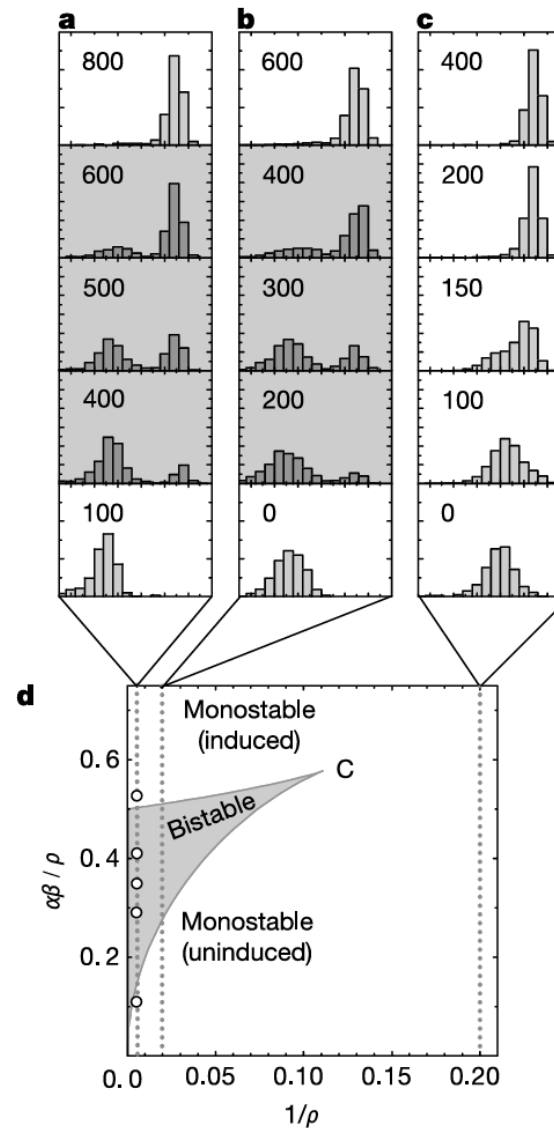
The model parameters correspond to measurable biological processes



The model predicts that weakening repression should destroy bistability



Reducing effective LacI repression converts the switch into a graded response



Graphical analysis reveals what is robust to microscopic details